THE THIRD ARROW OF A MORPHISM OF MAPPING CONES IS A HOMOTOPY EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3173-3201, following the statement at 3157-3171.

Editorial note. No omissions: both the reduction and the (1,1,c) case are fully displayed.

Allowed standard prerequisites (editorial).

- Functoriality of the cone construction. lemma-functorial-cone (same chapter, cited at ~3180). Builds the reverse morphism c' from the identities a^{-1}, b^{-1} in K(A).
- Nilpotence from termwise split sequences. lemma-nilpotent (same chapter, cited at ~3198). Gives (c - 1)^2 = 0, hence c is invertible with inverse 2 - c.

Source reference. Original lemma 014F: sources/stacks-derived/pinned-original-source.tex, source lines 2718–2718. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 086L: sources/stacks-derived/pinned-original-source.tex, source lines 3120–3120. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an additive category. Let $f_1 : K_1^\bullet \to L_1^\bullet$ and $f_2 : K_2^\bullet \to L_2^\bullet$ be morphisms of complexes. Let*

$$(a, b, c) : (K_1^\bullet, L_1^\bullet, C(f_1)^\bullet, f_1, i_1, p_1) \longrightarrow (K_2^\bullet, L_2^\bullet, C(f_2)^\bullet, f_2, i_2, p_2)$$

be any morphism of triangles of $K(\mathcal{A})$. If a and b are homotopy equivalences then so is c.

Proof. Let $a^{-1} : K_2^\bullet \to K_1^\bullet$ be a morphism of complexes which is inverse to a in $K(\mathcal{A})$. Let $b^{-1} : L_2^\bullet \to L_1^\bullet$ be a morphism of complexes which is inverse to b in $K(\mathcal{A})$. Let $c' : C(f_2)^\bullet \to C(f_1)^\bullet$ be the morphism from Lemma 014F applied to $f_1 \circ a^{-1} = b^{-1} \circ f_2$. If we can show that $c \circ c'$ and $c' \circ c$ are isomorphisms in $K(\mathcal{A})$ then we win. Hence it suffices to prove the following: Given a morphism of

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triangles $(1, 1, c) : (K^\bullet, L^\bullet, C(f)^\bullet, f, i, p)$ in $K(\mathcal{A})$ the morphism c is an isomorphism in $K(\mathcal{A})$. By assumption the two squares in the diagram

$$\begin{array}{ccccc} L^\bullet & \longrightarrow & C(f)^\bullet & \longrightarrow & K^\bullet[1] \\ \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle c} & & \downarrow{\scriptstyle 1} \\ L^\bullet & \longrightarrow & C(f)^\bullet & \longrightarrow & K^\bullet[1] \end{array}$$

commute up to homotopy. By construction of $C(f)^\bullet$ the rows form termwise split sequences of complexes. Thus we see that $(c - 1)^2 = 0$ in $K(\mathcal{A})$ by Lemma 086L. Hence c is an isomorphism in $K(\mathcal{A})$ with inverse $2 - c$. $\square$